Database [E22-CN3]
Term project

Overview

In this project, students need to design and build a database system for a predefined topic. Your database must be implemented in one of familiar database management system, such as Oracle, MySQL, SQL Server... This project must be done in groups of two or three students. Any changes in the group composition, once the groups have been formed will need my approval.

Completion of the project requires submission of different reports at various due dates throughout the semester as well as in-class system demonstration.

The project

Phase	Description	Due date	Points
1	Group formation	20/8/2024	0
2	"real-world scenario" description, Conceptual design, Logical design and Normalization	14/10/2024	40
3	Data implementation Final report	18/11/2024	20
4	Demonstration of completed system	26/11/2024	40

Phase 2:

In this phase, you need to identify the "real-world scenario" that you are assigned to model in your database, to construct an ERD, to convert the ERD into a set of relational schemas, and finally to normalize all the schemas. In other words, you have to implement steps 1, 2, 3, 4 of the database design process introduced in the class.
The report of this phase must include the following:

- All required information for the database that you are assigned, including a description of the database applications, an analysis of the data requirements which should include all of the relevant information that needs to be collected, all types of applications that need to be supported (loads, updates, retrievals, reports, …)
- An entity-relationship diagram (ERD) depicting your conceptual design. The ERD must capture all of the constraints that are possible using the E-R modeling concepts and notations. Any constraints or requirements of the business that cannot be modeled by the E-R notation should be clearly stated in English. Your ERD must present all types entity sets, all types of relationships.

- A set of relational schemas converted from the ERD presented above. All the primary key / foreign key connections among the relational schemas must be clearly presented.
- Normalization of the relational schemas. All resulting normalized relational database schema must be clearly presented.

Phase 3:

This phase deals with the implementation of the database using any type of database management systems. The database will be populated with sufficient amount of data to generate meaningful results for the various queries and applications that are also constructed during this phase.

Successful completion of this phase requires the submission of your entire database implementation. All data files and applications should be included with this submission.

The report in this phase should include all the details of your term project. It will be a compilation of the report you have submitted in the earlier phase, and the report of database implementation.

Phase 4:

During this phase your group will demonstrate your database system in class.

Requirements:

1. All members of the group must show up at the demonstration.
2. Every member of the group must present some part of the project.
3. Depending on your workload, your presentation time will be long or short.

General requirements

Each group will assign one member to send reports at corresponding due dates to my email, hoand@ptit.edu.vn . All email titles **must** include [E22-CN3] and your Group Numbers, **for example: [E22-CN3] group 2 – report phase 2**.

All reports are prepared in English, and must be submitted in one single file each time (MS Word, or PDF only). All other types of report will not be accepted.

Reports must be submitted by **23h59' on due dates**.

Term project will be graded in a 10 score scale. All members in each group are expected to get the same grades. However, if the contribution to the project in the group is different among its members, the grades will be adjusted accordingly.